| | | | |
|---|---|---|---|
| 4 | ❖ Both source and listener move<br>❖ They move one behind the other<br>❖ Listener follows the source | a) Apparent frequency depends on the velocities of the source and the listener.<br>b) $v_S$ and $v_L$ become opposite to that in case-3. | $n' = \left(\frac{v + v_L}{v + v_s}\right) n$ |
| 5 | ❖ Source at rest<br>❖ Listener moves towards the source | a) Distance between source and listener decreases.<br>b) Apparent frequency is more than actual frequency.<br>c) $v_S = 0$ in case-1. | $n' = \left(\frac{v + v_L}{v}\right) n$ |
| 6 | ❖ Source at rest<br>❖ Listener moves away from the source | a) Distance between source and listener increases.<br>b) Apparent frequency is less than actual frequency.<br>c) $v_S = 0$ in case-2. | $n' = \left(\frac{v - v_L}{v}\right) n$ |
| 7 | ❖ Listener at rest<br>❖ Source moves towards the listener | a) Distance between source and listener decreases.<br>b) Apparent frequency is more than actual frequency.<br>c) $v_L = 0$ in case-1. | $n' = \left(\frac{v}{v - v_s}\right) n$ |
| 8 | ❖ Listener at rest<br>❖ Source moves away from the listener | a) Distance between source and listener increases.<br>b) Apparent frequency is less than actual frequency.<br>c) $v_L = 0$ in case-2. | $n' = \left(\frac{v}{v + v_s}\right) n$ |

Suppose the medium (say wind) is moving with a velocity W in the direction of the propagation of sound. For this case, the velocity of sound, 'v' should be replaced with (v + W). If the medium moves in a direction opposite to the propagation of sound, then 'v' should be replaced with (v – W).

## Solved problems

1. A source producing a sound of frequency 90 Hz is approaching a stationary listener with a speed equal to (1/10) of the speed of sound. What will be the frequency heard by the listener?

**Solution:** When the source is moving towards the stationary listener, the expression for apparent frequency is

$$n' = \left(\frac{v}{v - v_s}\right) n$$

$$= \left(\frac{v}{v - \left(\frac{1}{10}\right) v}\right) n = \left(\frac{10}{9}\right) n$$

$$= \left(\frac{10}{9}\right) \times 90 = 100 \text{ Hz}$$

2. A source producing a sound of frequency 500 Hz is moving towards a listener with a velocity of 30 $ms^{-1}$. The speed of the sound is 330 $ms^{-1}$. What will be the frequency heard by listener?



10th_Science_Unit-5.indd 67 11-01-2020 12:20:12